

Institutional Trading Behavior in the ETF Market

Hsuan-Chi Chen
Anderson School of Management
University of New Mexico
Albuquerque, NM, USA
chen@mgt.unm.edu

Jen-Kai Ho
Department of Finance
Yuan Ze University, Taiwan
s957228@mail.yzu.edu.tw

Christine W. Lai
Department of Finance
Yuan Ze University, Taiwan
lai@saturn.yzu.edu.tw

Emmanuel Morales-Camargo[♦]
Anderson School of Management
University of New Mexico
Albuquerque, NM, USA
emmanuel@unm.edu

This Draft: 8/10/2011

[♦] Corresponding author. Please address all mail correspondence to Emmanuel Morales-Camargo, MSC05 3090, 1 University of New Mexico, Albuquerque, NM 87131-0001; Email: emmanuel@unm.edu; Phone: (505) 277-3403; Fax: (505) 277-7108.

Institutional Trading Behavior in the ETF Market

Abstract

We examine the trading behavior of institutional investors in the Exchange Traded Fund (ETF) market from 1993 to 2007. We concentrate on the relation between cross-sectional institutional ETF ownership and returns, particularly on the relation between changes in ownership and future returns, between past returns and institutional ownership, and between changes in ownership across institution classes (i.e. herding). Our results do not show evidence of a significant relation between changes in institutional ownership and future ETF returns. However, consistent with the use of negative feedback investment strategies, institutional investors tend to exhibit excess demands for past losers, especially in sector and international ETFs. They also exhibit high degrees of herding behavior, with banks driving the bulk of these results. Furthermore, their herding behavior is influenced by their experience in the ETF market and by certain ETF characteristics, such as liquidity and size, suggesting an aversion for information uncertainty leads them to herd.

Key words: ETFs, institutional investor behavior, institutional ownership, herding, feedback investment strategies

JEL Classification Codes: G10, G11, G14, G20

1. Introduction

From the first ETF traded on the Amex in 1993, the number of ETFs listed nationwide increased to 629 by the end of 2007. Similarly, the market value of the assets under management by ETFs grew from less than \$1 billion to over \$608 billion during the same period. In spite of this momentous growth, many important aspects of the ETF market remain relatively unexplored. A case in point is institutional investor behavior: While institutional investor behavior in the stock and mutual funds markets has been extensively studied, the trading behavior of this important investor group within the ETF market is yet to be explored. And while it cannot be completely ruled out *ex ante* that similar behavioral patterns may obtain, the significant institutional differences between the ETF and the other two markets suggest that important differences in motivation and trading behavior may exist. Understanding what drives institutional investors to trade ETFs may hold the key to figure out what is behind such impressive growth rates and the direction the ETF market will take in the future. Do institutional investors regard ETFs as just another financial instrument that they can use to exploit their private information (or speculate) about specific market sectors or do they actually use ETF order flow to infer information about the market? Do institutions just see ETFs as a simple and inexpensive means of diversification or do they rather use them as means to comply with some of their portfolio's investment policy guidelines? This paper fills an important gap in the financial literature by examining the behavior of various institutional investor classes in the ETF market and tries to tackle some of these important questions.

To be more specific, we concentrate on the relation between cross-sectional institutional ETF ownership and returns, particularly on the relation between changes in ownership and future ETF returns, and between past returns and institutional ownership. In addition, due to

the impact herding, or correlated trading amongst institutional investors, can have on these trading-return relations (Nofsinger and Sias (1999)), we also explore the prevalence of this phenomenon in this market and its impact on the aforementioned relations.

While we are aware that we cannot observe institutional investors' information set and cannot know for sure if they have private information, we are interested in seeing if there is evidence that institutional investors use ETFs directly to successfully speculate, as it has been documented with stocks and mutual funds in the financial literature. In other words, do institutional investors tend to build their ETF positions prior to an upturn in that ETF market category and reduce them prior to a downturn? To ascertain this, we evaluate the impact of changes in quarterly institutional ETF ownership on next-month ETF returns for four broad institutional investor classes (banks, insurance companies, investment companies, and independent investment advisors), using Pearson correlation coefficients and the Grinblatt and Titman's (1993) performance measure. Our tests with both measures reject the notion that institutional investors are able to strategically take positions in ETFs to significantly profit from them, rejecting the idea that they successfully use their private information (if they have it) to extract abnormal returns from the ETF market directly. Moreover, the negative Grinblatt and Titman performance estimates documented for banks suggest that these investors may use ETFs to fulfill other objectives, like complying with their portfolio investment policy guidelines or conforming to the industry norm.

To examine whether past ETF performance has an impact on institutional investor demand for ETF shares, namely whether institutional investors use feedback investment strategies, we construct excess ETF demand measures similar to those used within the context of the stock market by Lakonishok, et al. (1992). In contrast with most of the literature that studies

institutional investors' use of feedback investment trading strategies within the context of the stock market, our tests find that institutions engage in negative, not positive investment strategies (Lakonishok et al. (1992), Grinblatt, et al. (1995), Nofsinger and Sias (1999), Badrinath and Wahal (2002)). Also in contrast with Lakonishok et al. (1992), we do not find that institutional investors exclusively feedback-trade on small size securities; they actually contrarian-trade using both small and large market capitalization ETFs. These findings suggest that institutional investors see the ETFs market as a viable means to trade on the basis of potential market overreaction to new information. In addition, our examinations uncovers some selectivity across ETF types in the application of these contrarian strategies. Excess demand for past losers seems to be concentrated on U.S. sector and international ETFs, while excess supply of past winners seems concentrated on U.S. Broad and Style ETFs. These last results may be an indication that institutional investors tend to use some specific ETF types for the purpose of diversification.

In order to determine whether herding behavior in the ETF market is a pervasive phenomenon, we employ the herding measure developed by Lakonishok, et al. (1992). We find that herding is significant in ETFs with high levels of bank involvement. In contrast, amongst independent investment advisors, herding is only significant in the style ETF market when the sample is restricted to ETFs with high levels of advisor involvement.

Further regression analysis shows that herding behavior is affected by specific ETF traits and by the institutional investors' experience within the ETF market, rather than by potentially correlated private information amongst investors. Since herding is most pervasive within relatively illiquid and small ETFs, it is likely that institutional investors have a particular aversion for information uncertainty, regarding any trade in these ETFs as a sign of

new information arrival and following with similar trades. As for the role of experience in herding behavior, the regression results suggests that herding takes place mostly among institutional investors who are relatively inexperienced in the ETF market.

The remainder of this paper is organized as follows. The next section reviews the literature on institutional trading behavior in the stock and mutual funds markets and motivates the study. Section 3 describes the datasets used and the sample in our tests. Section 4 describes the construction of the different measures used in our tests. Section 5 presents the empirical results and Section 6 concludes.

2. Literature Review

In the past, the finance literature has examined herding behavior and the relation between institutional investors' trading behavior and returns within the context of the stock and mutual fund markets. However, institutional investor behavior has not yet been explored within the context of the ETF market in spite of the fact that significant institutional differences from traditional mutual funds may lead to significant differences in investors' motivation and trading behavior. For instance, unlike with mutual funds and stocks, ETF managers do not have the explicit objective of providing abnormally high returns to their shareholders; they just track an index. Consequently, a traditional ETF investor will only see abnormally high returns if the index the ETF tracks happens to have a superior performance. Does that mean that institutional ETF investors are not actively looking to directly profit from their ETF holdings, but are rather investing in ETFs to achieve other objectives within their portfolios? Are ETFs perceived only to be a low-cost diversification alternative? Are institutional investors just interested in a low-cost medium to meet their asset allocation/sector rotation

targets, to comply with their investment policy guidelines, or to conform to the industry norm? By studying the correlation between changes in institutional investor ETF holdings and ensuing ETF returns we are able to shed some light on some of these questions.

A positive correlation between cross sectional changes in institutional ownership and future ETF returns can be explained several ways. One possibility is that institutional investors possess private information or superior insight (i.e. timing ability) about the market, allowing them to increase (decrease) their position in ETFs before an upturn (downturn). Naturally, for this scenario to hold, institutional investors must have market timing ability in the first place and regard ETFs as optimal instruments to carry out their investment strategies, preferring them over (or perhaps use them in combination with) individual stocks, bonds, and other derivative instruments. Alternatively, if institutional trading carries enough volume, prices may just move in response to price pressure, going up as institutions build up their position in an ETF and going down as they divest from it. While the existence of market timing abilities or superior information would not be necessary under this last scenario, it is possible that the forces in both scenarios reinforce each other. Since the price-pressure explanation is more likely to make sense in the style and sector ETF than in the larger market capitalization categories of the U.S. broad and international ETF markets, the relative magnitude of the correlation coefficient could potentially help us discriminate between these two competing theories.

On the other hand, a negative correlation between cross sectional changes in institutional ownership and future ETF returns is also possible. Naturally, if institutional investors time the market unsuccessfully, they may build up their ETF positions before a downturn and liquidate them before an upturn, losing a lot of money in the process. However, a negative correlation

may also obtain if we consider some of the alternate purposes ETFs can serve in institutionally managed portfolios. For instance, many institutional investors manage portfolios that are subject to investment policies that require a long exposure to an asset class, a style, a sector, or even a foreign country. While actively managed portfolios would typically comply with those requirements using individual securities (i.e. stocks and bonds), when managers expect some of their individual securities to tumble and do not have substitute securities in that market category, many drop these individual securities in favor of ETF shares (i.e. a diversified position) in the market category in which they must maintain a particular exposure. Thus, they may end up loading up on ETFs ahead of a downturn and divesting from ETFs ahead of upturns. While those measures do not insulate institutional investors from losses, they arguably help institutions minimize them and comply with their investment policy guidelines.

Naturally, within the context of this last hypothesis, evidence of a negative correlation between changes in ETF holding and future ETF returns would necessarily imply that institutional investors possess some degree of market timing ability in stocks, bonds, or other individual securities (i.e. the individual securities they are selling and that end up moving in the expected direction along with others in their market group). However, ETFs are not the main means through which the institutions capitalize on their market timing abilities, but rather a transient tool. This scenario could also hold if some institutional fund managers, out of job security considerations, try to maintain some asset class or sector targets at par with their industry peers, as it has been documented for younger and relatively inexperienced mutual fund managers in the financial literature (Chevalier and Ellison (1999)).

Interestingly, regardless of the direction of the correlation between changes in institutional

ETF holdings and future ETF returns, pervasive herding behavior can help magnify or even explain the impact of institutional trading, making it necessary to jointly study this type of behavior when evaluating the aforementioned correlation.

For the most part, the relation between changes in institutional holdings and ensuing returns has been studied empirically from the market timing standpoint with mutual funds, but has not yielded consistent results. Bollen and Busse (2001) and Chance and Hemler (2001), for instance, using daily mutual fund data find evidence supporting the market timing ability of mutual fund managers and investment advisers, respectively. Chakravarty (2001) shows that medium-size trades from institutional investors are associated with large cumulative stock price changes, suggesting that institutions stealth trade whenever they exploit their private information. However, Grinblatt and Titman (1993), using a 1-quarter lag measure, find no market timing ability amongst mutual fund managers. While, they are able to find some market timing ability using a 4-quarter lag measure, they only do so before accounting for mutual fund fees. Moreover, Daniel et al. (1997), using a characteristic-based performance measure, find no evidence in favor of mutual fund managers' market timing ability.¹

Given the mixed evidence uncovered within the context of mutual funds, whether other types of institutional investor have any market timing ability is naturally still an open empirical question. However, we recognize that, in contrast with prior tests of mutual fund managers' abilities, any tests that look exclusively at changes in institutional ETF holdings in search for proof of institutional investors' market timing ability have an inherent problem: they ignore all the other instruments institutional investors could use to profit from their

¹ An alternative way to evaluate if institutional investors have market timing ability is to look for smart money effects. The smart money hypothesis states that investors' money finds its way to the funds that will outperform in the future, and thus, implies that investors have genuine fund selection and market timing abilities. However, evidence of the existence of smart money within the mutual funds market is also inconclusive.

timing abilities. Thus, while it is possible that institutional investors' ETF transactions turn out profitable, their other transactions may lose them money, leaving them with neutral or even negative returns and disproving their market timing ability. Similarly, while the investors' ETF transactions may not yield any positive returns, their other transaction may prove to be very lucrative, leaving them with abnormal returns that would evidence their timing skills.

In light of these measurement problem, it is not our intentions to prove whether the institutional investors in our sample posses any market timing abilities, private information, or superior analytical insight. Rather, we focus on institutional investors' observable behavior in the ETF market and try to answer a more unassuming question: Is there factual evidence that institutional investors are able to successfully take strategic positions in the ETF market and directly profit from them? We acknowledge that a negative answer to this question may suggest institutional investors either have no market timing ability or that they chose not to exploit it with ETFs in a straightforward manner (i.e. just buying ETFs before an upturn or selling them before a downturn). Regardless of which one may be, this result would suggest that institution must trade ETFs to fulfill other objectives, a realization that would help us move our understanding of the ETF market one step forward.

Moving on to the second relation of interest in this study, the financial literature has ample documented evidence supporting institutional investors' use of feedback investment strategies within the stock market. Most studies, however, find that institutional investors are positive feedback (momentum) traders (Lakonishok et al. (1992), Grinblatt, et al. (1995), Nofsinger and Sias (1999), Badrinath and Wahal (2002)) and that they are highly selective about which instruments they trade in and when they apply these trading strategies. Lakonishok et al. (1992) for instance, find that pension funds pension funds only seem to use positive feedback

strategies when dealing with small stocks, Grinblatt, et al. (1995) find that institutions only buy past winners, and Badrinath and Wahal (2002) find that institutions act as momentum traders when they enter stocks, but as contrarian traders when they exit or make trim down their current holdings. While we are not certain what to expect about institutional investors' behavior in the ETF market when it comes to the use of feedback investment strategies, we suspect we will encounter similar evidence in our sample.

We recognize however that the use of feedback investment strategies may also be partially explained by herding behavior as institutional investors flock together to purchase or sell ETFs based on their past performance. Lakonishok, et al. (1992), for instance, find that while institutional investors use inconsistent strategies when trading stocks, they usually flock together. Wermers (1999) also finds some evidence of herding by mutual funds for the average stock and strong evidence of herding in small stocks and by growth-oriented funds. Therefore, we also expect to find some evidence of herding behavior by institutional investors when they trade ETFs and we are interested in finding out how that phenomenon affects the correlation between institutional investment changes and ETF returns.

3. Data

Institutional shareholding data and ETF monthly return data come from CDA/Spectrum and the Center for Research in Security Prices (CRSP), respectively. The sample is comprised of all the ETFs appearing in the CRSP database as of December 31, 2006. The sample period extends from January 1993 to December 2007. Since CRSP includes HOLDERS in the same share code as ETFs (share code 73), we exclude them from the sample. We address any survivorship bias concerns by not eliminating ETFs that were delisted during the sample

period. Consequently, the final sample is comprised of 342 active ETFs and 25 delisted ETFs.

CDA/Spectrum Institutional (13f) Holdings classifies institutional investors into five groups: (1) banks, (2) insurance companies, (3) investment companies and their managers, (4) independent investment advisors, and (5) all others. Since CDA/Spectrum does not provide a detailed description of the fifth category, it is excluded from the sample. The independent investment advisor category, on the other hand, is well detailed and includes both capital management advisors and asset management advisors; thus, it is kept in the sample.

4. Methodology

We study the investment behavior of institutional investors in the ETF market, focusing more specifically on the relation between changes in institutional ETF holding and returns and the prevalence of herding behavior. We next describe how we measure these behavioral characteristics in more detail.

4.1. Impact of Institutional Trading on Future ETF Returns

If there is a significant relation between changes in institutional investor ETF holdings and ensuing ETF returns, it should be straightforward to find that changes in institutions' ETF investments or portfolio weights covary with ETF returns. Otherwise, the covariance between changes in their ETF investments and ETF returns should be zero. We evaluate the existence of this covariance in two ways, using both Pearson correlation coefficients and Grinblatt and Titman's (1993) measure.

Grinblatt and Titman's measure, an alternative way to compute the covariance, is adapted for the purposes of this study as follows:

$$ETF\ Change\ Measure_i = \sum_{t=1}^T \sum_{j=1}^N [R_{j,t} (w_{j,t}^i - w_{j,t-1}^i)] / T \quad (1)$$

where $R_{j,t}$ = ETF j 's return in the month immediately following Quarter t ,

$w_{j,t}^i$ = the weight of institutional investor i 's position in ETF j in Quarter t .

Under the null hypothesis that institutional investor i 's change in ETF holdings has no effect in future ETF returns, we expect this performance estimate to be indistinguishable from zero.

4.2. Feedback trading strategies

To examine whether past ETF performance has an impact on institutional investor demand for ETF shares, we first construct two excess ETF demand measures similar to those used within the context of the stock market by Lakonishok, et al. (1992). One measure looks at dollar excess demand, *Dratio* (dollar ratio), and another one looks at the number of institutional investors increasing their ETF holdings during the quarter, *Nratio* (number ratio). Within the context of this study, *Dratio* _{j,t} is defined as:

$$Dratio_{j,t} = [\$buys_{j,t} - \$sells_{j,t}] / [\$buys_{j,t} + \$sells_{j,t}] \quad (2)$$

where

$\$buys_{j,t}$ = the total dollar value of increases in ETF j holdings by all institutional investors in Quarter t (evaluated at the average ETF price during the quarter)

$\$sells_{j,t}$ = the total dollar value of decreases in ETF j holdings by all institutional investors in Quarter t (evaluated at the average ETF price during the quarter)

and *Nratio* is defined as:

$$Nratio_{j,t} = \#buys_{j,t} / \#active_{j,t} \quad (3)$$

where

$\#buys_{j,t}$ = the number of institutional investors increasing their holdings of ETF j in
Quarter t

$\#active_{j,t}$ = the number of institutional investors changing their holdings of ETF j in
Quarter t

The interpretation of these two excess demand measures is straightforward: The larger their value, the more shares of ETF j institutional investors have opted to buy in Quarter t . If institutional investors use a positive feedback investment strategy based on past ETF performance, one would expect that whenever an ETF performed well (poorly) in the previous quarter, the *Dratio* and *Nratio* would be larger (smaller) in the current quarter. Similarly, when institutional investors use a negative feedback strategy, good (poor) ETF performance in the previous quarter would result in smaller (larger) *Dratios* and *Nratios* in the current quarter. To test this conjecture, we compare the average *Dratio* and average *Nratio* of ETFs across previous-quarter performance quintiles, dividing the sample into ETF size (market capitalization) quintiles to control for size effects. The use of size controls follows the work of Lakonishok, et al. (1992), who find that pension funds only seem to use positive feedback strategies when dealing with small stock; a reflection perhaps of the relatively large informational asymmetries that prevail amongst this stock group.

4.3. Herding

We study institutional investor herding behavior by testing whether institutions tend to flock on the same side of the market of a particular ETF in a given quarter. That is, we look at whether a disproportionate number of the institutional investors are purchasing (or selling) a particular ETF in any given quarter, using the measure introduced by Lakonishok, et al.

(1992). When a large percentage of institutional investors (let's say 60%) end up on the same side of the market, we conclude that there is herding in that particular ETF.

The herding measure for ETF j in Quarter t is defined as:

$$H_{j,t} = \left| \frac{B_{j,t}}{B_{j,t} + S_{j,t}} - P_{j,t} \right| - AF_{j,t} \quad (4)$$

where $B_{j,t}$ = the number of institutional investors who increase their holdings in ETF j in Quarter t (net buyers),

$S_{j,t}$ = the number of institutional investors who decrease their holdings in ETF j in Quarter t (net sellers),

$P_{j,t}$ = the expected proportion of institutional investors buying ETF j in Quarter t relative

to the number of active institutional investors in that ETF that quarter,

$AF_{j,t}$ = an adjustment factor that equals the expected value of the absolute term in (3), under the null hypothesis of no herding.

The ratio in Equation (4), computed by dividing the number of net buyers by the total number of active institutional investors in that ETF that quarter, measures the proportion of buyers in a given ETF in a given quarter. However, this ratio may be biased by the buying atmosphere in the market in any given quarter, resulting in a high ratio whenever the market is bullish. Thus, the main idea behind the use of adjustment factor $AF_{j,t}$ in Equation (4) is to correct for this bias. As Lakonishok et al. (1992) explain, since $B_{j,t}$ in Equation (4) follows a binomial distribution with known probability p of buying, $AF_{j,t}$ can be easily computed once the number of active institutional investors in ETF j that quarter is known. In the appendix, we provide a more detailed description and a numerical illustration of the computation process.

5. Empirical Results

Using AMEX and Morningstar taxonomy, the ETF market can be classified into ten ETF groups, according to their investment objective: (1) U.S. broad market, (2) style, (3) U.S. sector, (4) bond and fixed income, (5) commodity and currency, (6) dividend, (7) short, (8) international market, (9) global sector, and (10) others. The U.S. broad market category includes ETFs that track major U.S. market indices, like the S&P 500 and Dow Jones indices. The international market group does the same, but for foreign markets, including large capitalization markets like the London Stock Exchange and smaller but “hot” emerging markets like Brazil, Russia, India, and China. The style group mainly tracks U.S. growth, value, small-, middle-, and large-cap indices. The U.S. sector category includes ETFs that follow industry sector indices like health, energy, materials, and utilities indices. The global sector category does the same as this last group, but includes foreign securities within the different industry sectors, not just U.S. securities. The dividend group contains ETFs that try to replicate indices of high dividend yield stocks and ETFs in the short category try to provide the opposite performance as their benchmark index. While it should be obvious what type of indices the fourth and fifth categories follow, the “others” category contains only one ETF that reflect the performance of the U.S. initial public offerings market.

Figure 1 depicts the percentage of the overall ETF market capitalization each of these categories represents. The most popular ETF group is the U.S. sector, capturing 36% of the market, followed by the style group with 19%, the international market group with 13%, and the U.S. broad market group with 9%. Together, these four ETF groups represent about 77% of the overall ETF market.

[Insert Figure 1 Here]

Even though the U.S. broad market group had a head-start from the others with the inception of State Street Global Advisor's S&P 500 SPDR in 1993 (the first ETF), it is not currently the largest ETF group. Good performance in select U.S. sectors over the years has typically been followed by a series of ETFs tracking those "hot" sectors. In addition, it is more straightforward and often times more profitable to constitute a sector ETF that will track fewer securities than a broad market ETF that requires taking positions in hundreds of securities. Similarly, although the international market ETF group is a relatively new comer, it has grown significantly, particularly after 2006. Some scholars suspect this growth is probably due to the diversification advantage international market ETFs provide (Tse and Martinez (2007)). However, it is likely that the relative underperformance of the U.S. stock market in general has also contributed to this ETF group growth. In contrast, many of the groups in the bottom six categories, like the Short ETF group, probably owe their relatively low market capitalization precisely to their recent inception. Since those ETF groups do not provide us with sufficient observations through out this study's sample period, we focus our tests only on the top four ETF categories.

[Insert Figure 2 Here]

Figure 2 depicts the number of shares held by institutions in these four top ETF groups from 1993 to 2007. As this figure shows, institutional investors' share holdings in these ETF groups grew significantly after 2001. Thus, to account for regime shifts in the evolution of institutional investors' stake in the ETF groups, we divide the sample period into two sub-periods: 1993 to 2000 and 2001 to 2007. As Figure 2 also shows, prior to 2006, the number of shares held by institutional investors in U.S. Broad Market ETFs exceeded their holdings in the other three ETF groups. The latter is true in spite of the lower number of ETFs

in the U.S. Broad Market category and may be the result of a first mover advantage of this group over the others.

Figure 3 reports the percentage stake held by institutional investors in the top four ETF groups from 1993 to 2007. The figure shows an accentuated upward trend after 2003 for all institutional investors types, except for insurance companies, who only began to invest heavily in ETFs after 2004. It also shows a significant plunge in ETF ownership by investment companies from mid-2004 to 2005, followed by a dramatic comeback in the second half of 2006. Independent investment advisors' stake also saw a similar plunge in 2006 that has not yet seen a reversal. The fact that institutional investors' percentage stakes in ETFs, for the most part, remains negligible until mid-2000 reinforces our earlier decision to break the sample into two sub-periods, using December 31, 2000 as the breakpoint.

[Insert Figure 3 Here]

Table 1 provides the ownership distribution of each ETF category across the four institutional investor types identified in the data section, using the sum of their stake in that ETF category as the basis for the percentage ownership figures reported. The table shows that, amongst these four institutional investor categories, throughout the 15-year sample period, independent investor advisors and banks hold majority stakes across all ETF categories, with a combined average ownership that ranges from 87% of the style ETF market to 92% of the U.S. sector and international ETF markets. While insurance companies' stake fluctuates between 8%, in the case of U.S. broad market ETFs, and 15%, in the case style ETFs, investment companies' stake is relatively small, fluctuating between zero and 4%, in the case of the U.S. sector and Style ETF markets, respectively. Moreover, between the two top investor types, independent investment advisors seems to be the majority shareholder in all

ETF types, except in style ETFs, where they hold a 33% stake while banks own 54% of the market.²

Table 1 also shows how relative share ownership in the ETF market has evolved over the two sub-sample periods. While banks percentage stake of the combined share ownership of the four institutional investor types increased from the first sub-sample period to the next in all ETF categories (as evidenced by both means and medians), the other investors seemed to decrease their ownership stake in the ETFs market over time. The only exceptions were the insurance companies' mean percentage stake in the international ETF market, which increased from 6% to 9%, and the independent investment advisors' mean (median) percentage stake in the U.S. broad market category, which increased from 48% to 53% (47% to 56%).

[Insert Table 1 Here]

5.1. Impact of Institutional Trading on Future ETF Returns

We evaluate the impact of institutional investors' trades on future ETF returns by first computing Pearson correlation coefficients between the change in their quarterly ETF investment weights and the corresponding ETF's return the following month. If institutional investors successfully take strategic positions in ETFs to profit from movements in that ETF group, or if they exert sufficient price pressure to move prices, the correlation between these two variables should be significantly positive. If institutional investors fail to profitably take strategic positions in ETFs or use ETF merely as transient investments when executing their investment strategies with stocks, bonds, or any other individual financial instruments, then the correlation should be negative.

Based on the Pearson correlation figures in Table 2, it seems that only changes in bank weights have a significantly positive correlation with future ETF returns during the 15-year

² Median figures in Table 1 lead to qualitatively similar conclusions.

sample period. Moreover, that is only true within the style and international ETF markets. Given banks' relative importance in the Style ETF market, the price pressure hypothesis seems like a reasonable explanation for these results. Price pressure also seems sensible to explain the observed results in international ETFs, provided the market capitalization of the international markets banks invest in is relatively small.

However, it is pertinent to note that, although statistically significant, the correlation between banks' ETF trades and the ensuing returns could hardly be considered economically significant. With a correlation coefficient of only 0.003 (See Table 2B), even sizable increases in banks' international ETF quarterly ownership are followed by negligible return upturns. Assume for instance that a bank increases its investment in iShares' Mexican Market Index (EWW) from 39% to 42.9%. That increase represents a relative jump of 10% from the average percentage stake held by banks in international ETFs during the sample period (See Table 1A). However, if we consider that from its inception to the end of 2007, EWW had an average monthly return of 1.32%, such a bold move would only be followed by an average 0.04 basis points (BPS) increase in monthly returns. In sum, banks do not seem to be able to strategically take positions in ETFs to profit significantly from them.

Contrary to banks, independent investment advisors' international ETF trades turn out to be negatively correlated to future ETF returns. Such a loading conflicts with the price pressure hypothesis and with the notion that investment advisors are able to take strategic positions in international ETFs for a profit. Since it is unlikely that the international ETF market would have experienced such a dramatic growth over the past few years at the same time that investors trying to speculate on it kept losing money, we discount a failed investment strategy as the potential explanation for this negative correlation. Rather, this correlation

seems to support the notion that international ETFs serve as a transient venue for independent investment advisors when their investment policy guidelines require them to keep a long exposure to international markets.

[Insert Table 2 Here]

To examine the relation between institutional trades and future ETF returns in an alternative way, we also construct Grinblatt and Titman's (1993) performance estimates for each institutional investor type, as outlined in Section 4.1. Table 3 presents the mean of these ETF change (performance) measures for the top four ETF groups and their respective t-statistics.³ As Panel A shows, these performance measures are not supportive of a significantly positive relation between institutional investor trading and future ETF returns. In fact, they are significantly negative, both in the statistical (see t-statistics) and economic sense. For the full sample period, for instance, institutional investors' changes in quarterly weights have been followed by negative monthly returns that range from 11 BPS, in the case of U.S. broad market ETFs, to 20 BPS in the case of U.S. sector ETFs.

Moreover, as Panel B through E show, these results are mostly driven by banks, whose performance is negative across ETF types and across the two sub-sample periods. Insurance companies also contribute somewhat to the overall negative performance of institutional investors, with a significantly negative performance for the style ETF group in the full and second half of the sample period, and for the U.S. Sector ETF group in the first half of the sample period. Once again, these negative performance measures seem to support the notion that international ETFs serve as a transient investment venue for institutions when required to keep a long exposure to an ETF group by their portfolios' investment policy guidelines (or

³ We also use Wilcoxon's non-parametric test to account for the potential non-normality of ETF stake holding data and for the small sample size within the insurance and investing company sub-samples. The results of these tests support the inference in Table 3 using t-statistics and are not tabulated.

industry norm), rather than as a direct investment venue they seek to profit from.⁴

[Insert Table 3 Here]

5.2. Feedback trading strategies

To measure the extent to which institutional investors use feedback trading strategies, we study the correlation between past ETF performance and excess institutional investor demand within that market. In a similar vein, we look at the correlation between past ETF performance and the proportion of institutions building positions in the ETF market. When institutional investors purchase (sell) ETFs with good (poor) performance the previous quarter, they are considered to follow a positive feedback strategy. When they purchase (sell) ETFs with poor (good) past performance, they are considered to follow a negative feedback (or contrarian) strategy.

Table 4 reports the quarterly excess demand ratio, *Dratio*, and the proportion of institutions buying ETF shares, *Nratio*, averaged across ETF size quintiles and quintiles formed on the basis of past quarterly performance. We divide the sample into ETF size (market capitalization) quintiles to control for size effects in response to the work of Lakinishok, et al. (1992), who find that evidence supporting the use of positive feedback strategies by a group of pension funds disappears as stock size increases. Consequently, the table reports *Dratios* and *Nratios* for a total of 25 ETF cohorts. Once again, we limit the sample to the period following 2001 to ensure that each ETF cohort is comprised of at least five ETFs. In addition, to avoid having our measures biased by one or two institutional investors, we require all ETFs in each size-performance cohort to be traded by at least five institutional investors.

⁴ Although not tabulated, qualitatively identical results to those in Table 3 were obtained using the Characteristic Timing measure developed by Daniel, et al. (1999). We did that to account for the possibility that institutional investor's failed attempts to take strategic positions in ETFs may be due to ETF tracking error, rather than their own abilities.

[Insert Table 4 Here]

As Table 4A shows, excess demand is significantly positive for six ETF cohorts within the two worst performance quintiles and significantly negative for three ETF cohorts within the two best performance quintiles. This evidence suggests that institutional investors are not momentum traders, but rather follow a negative feedback investment strategy, buying past losers and selling past winners.

Moreover, the evidence in Table 4A shows that institutional investors' use of negative feedback investment strategies is not a practice limited to the small ETF market. This conclusion is supported by the fact that, while there are four cohorts with significantly positive excess demand at the intersection of the two worst performance cohorts and the two smallest size cohorts, there are two cohorts with significantly positive excess demand at the intersection of the two worst performance cohorts and the two largest size cohorts. By the same token, there are three cohorts with significantly negative excess demand at the intersection of the two best performance quintiles and the two largest size quintiles.

Although not tabulated, it is worth mentioning that about 75% of the ETFs in the four small-size cohorts with significantly positive excess demand belong to the U.S. sector and international ETF categories. On the other hand, about 68% of the ETFs in the two cohorts with significantly negative excess demand within the second best performance quintile are U.S. broad and style ETFs. If one takes into account the fact that one of the top performing ETF groups during the sampled period is the international ETF group, with about half of the ETFs in the second to best performance quintile belonging to this ETF category, it would appear that the use of negative feedback trading strategies within the ETF market is somewhat selective and may reflect diversification motives.

The above conclusions concerning the use of negative feedback investment strategies seem to be corroborated by the *Nratio* measures in Panel B. That is, the ETF cohorts within the two worst performing quintiles that exhibit significantly positive excess demand appear to be more aggressively bought (as evidenced by their relatively high *Nratio* measure) and the three ETF cohorts within the two best performing quintiles that exhibit significantly negative excess demand appear to be more passively bought.

5.3. Herding

In the financial literature, there is empirical and survey evidence (Friend, et al. (1970), Schiller and Pond (1989), and Lakonishok, et al. (1992)) supporting the notion of herding behavior in the stock market amongst some institutional investors. In order to determine if herding behavior in the ETF market is pervasive, we employ the herding measure developed by Lakonishok, et al. (1992). We focus strictly on the 2001-2007 sample period because prior to 2001 institutional investor presence in the emerging ETF market was still relatively low, and many ETFs were only held by less than a handful of institutional investors. In total, we are able to compute herding measures for 2,779 ETF-quarters.

The first column of Table 5 reports mean and median statistics of herding behavior within all the ETFs in the sub-sample, regardless of the number of institutional investors actively trading (i.e., buying and selling) the ETFs. The next two columns focus only on herding behavior within ETFs with at least five and ten active institutional investors, respectively. Column 4 in Panels B through D reports the difference between the herding measures in Columns 3 and 1.

As mean herding measures in Panel A show, herding amongst institutions seems pervasive when we examine ETFs with active institutional investor involvement. That is, while no

herding is detected when all ETFs are examined, herding becomes apparent once we restrict our measurements to ETFs with at least five or ten active institutional investors.

[Insert Table 5 Here]

While Panel B shows a similar pattern for all the ETF categories of interest in this study, Panels C and D suggest that these results are mostly driven by banks. As Panel C shows, when we place no restrictions in the level of bank involvement in any of the ETFs categories, no herding appears to take place.⁵ However, once we restrict the sample to ETFs with high levels of bank involvement, we see that herding becomes apparent. In contrast, amongst independent investment advisors, once we restrict the sample to ETFs with high level of advisor involvement, herding only becomes apparent in the Style ETF market. No evidence of herding amongst independent investment advisors surfaces in the other three ETF categories.⁶

Although our tests are unable to pinpoint the exact reason why herding amongst banks in the ETF market seems to be more pervasive than amongst independent investment advisors, this phenomenon may be partially due to the geographical proximity of banks' trading offices, which may facilitate the dissemination of trading information within the banking community. That is, while all major banks have trading offices in all the major financial hubs around the globe, independent investment advisors are more geographically scattered and may remit their trades through more diverse and stealth channels.

5.4. Determinants of Herding Behavior

⁵ The reader should not put too much weight on the significantly negative sign of the herding measures reported in Column 1 of Panels C and D. Rather than indicative of any pervasive trading behavior, these negative numbers are the product of the high value the adjustment factor, $AF(i)$, takes in Equation (3) when the number of institutional investors actively trading an ETF is low. See the Appendix for further details on the computation of $AF_{j,t}$.

⁶ Due to the low number of insurance and investment companies actively trading in the ETF market throughout the sample period (see Panels C and D in Table 3), it is not possible to analyze herding this way for these two types of institutional investors.

Since the previous section presents evidence supportive of pervasive institutional investor herding behavior in the ETF market, we now turn our focus to the driving factors behind this behavior. Through regression analysis, we evaluate three distinct, although not necessarily mutually exclusive hypotheses explaining herding: (i) correlated information, (ii) common preference/aversion to certain ETF characteristics, and (iii) institutional investors experience in the ETF market. We explain the nature of these hypotheses, the methodology used to test them, and our test results in the remainder of this section.

5.4.1. Correlated Information

The herding behavior observed in this study could easily be explained by correlated information in the hands of institutional investors. That is, if institutional investors possess positively correlated information about an ETF's future return prospects, it would not be surprising to find multiple investors trading that ETF on the same side of the market; an event that can be construed as herding. Only those institutional investors with significantly dissimilar private information would be willing to take the other side of the market. Moreover, when information is costly, following other institutional investors' lead may be optimal, providing an efficiency rationale for herding in such an information environment.

We investigate whether the existence of (uncorrelated) private information within different institutional investor types (indexed by k) affects their herding behavior by regressing a measure of herding behavior for institutional investor type k in Quarter t ($HERDING_{k,t}$) on a modified version of the performance measure in Equation (1). We consider this measure an ETF market timing component of institutional investor type k in Quarter t ($SMART_MONEY_{k,t}$), and it is defined as follows:

$$SMART_MONEY_{k,t} = \sum_{i=1}^{I(k)} \sum_{j=1}^N [R_{j,t} (w_{j,t}^i - w_{j,t-1}^i)] / I(k) \quad (5)$$

where $R_{j,t}$ = ETF j 's return from Quarter t to Quarter $t + 1$,

$w_{j,t}^i$ = the weight of institutional investor i 's position in ETF j in Quarter t , and

$I(k)$ = Total number of institutional investors of type k in the sample.

The larger the measure in Equation (5), the more institutional investors adjust their ETF holdings ahead of positive performance, suggesting that they may be exploiting private information. Since we expect herding to decrease as the level of private information on institutional investors' hands increases, the coefficient of $SMART_MONEY_{k,t}$ in the aforementioned regression should be negative.

The dependent variable in this regression, $HERDING_{k,t}$, is computed by calculating a weighted average of the value of the herding measure in Equation (4), across all institutional investors in investor category k , using the percentage invested in ETF j as the weighting factor. That is,

$$HERDING_{k,t} = \sum_{i=1}^{I(k)} \sum_{j=1}^N w_{j,t}^i H_{j,t} \quad (6)$$

where $H_{j,t}$ = the herding measure for ETF j in Quarter t , as defined in Equation (4),

$w_{j,t}^i$ = the weight of institutional investor i 's position in ETF j in Quarter t ,

$I(k)$ = total number of institutional investors of type k in the sample.

5.4.2. Common Preference/Aversion for ETF Characteristics

According to studies surveying the preferences of mutual fund investors for different stock characteristics (Falkenstein, 1996), investors seem to exhibit a preference for certain

stock characteristics (like high volatility and high liquidity) and an aversion for others (like small market capitalization and high information uncertainty). To test whether any of those characteristics affect herding behavior within the ETF market, we add an ETF liquidity measure ($LIQ_{j,t}$) to the above regression, defined as trading volume in ETF j in Quarter t , divided by shares outstanding at the beginning of Quarter t . We also add as an independent variable to the regression the natural logarithm of ETF j 's market capitalization at the beginning of Quarter t ($LN_MKTSIZE_{j,t}$). Given Falkenstein's (1992) findings, we may see that the coefficients on LIQ and $LN_MKTSIZE$ turn out positive, suggesting that investors flock to purchase highly liquid and large ETFs. These results, however, would raise the possibility that they have a preference for low trading costs and high information transparency, resulting in trading behaviors that may be construed as herding in the ETFs that satisfy those criteria. However, if these two variables proxy for other ETF characteristics for which institutional investors exhibit an aversion, we could find the opposite results. Since small and less liquid ETFs may suffer from more information uncertainty (concerning the ETF's future return prospects), it could be that LIQ and $LN_MKTSIZE$ may turn out negative, implying that due to the uncertainty surrounding the fund, trades by one institutional investor are regarded by others as a signal of new information arrival, resulting in further trading in those ETFs form their part (i.e. herding).

5.4.3. *Institutional Investor Experience*

Herding behavior amongst institutional investors may also be influenced by their level of experience in the ETF market. As Chevalier and Ellison (1999) find, younger and relatively inexperienced mutual fund managers try not to deviate too much from the mean sector

weightings of their peer group or from the average level of systematic risk taken by mutual funds that share the same investment objective. They behave that way out of job security considerations, as Chevalier and Ellison (1999) find that those who deviate from the norm are often terminated. In other words, inexperienced managers tend to herd with their peers to avoid getting fired.

We test whether a similar phenomenon takes place amongst inexperienced institutional investors within the context of the ETF market. However, faced with the impossibility of obtaining institutional investor managers overall investment experience, we conduct our tests using the level of institutional investor experience in the ETF market, adding variable $EXPERIENCE_{i,t}$ to the above regression, where $EXPERIENCE_{i,t}$ is defined as the number of years institutional investor i has been trading in the ETF market up to the beginning of Quarter t . If our conjecture is correct, we expect to find a negative relation between dependent variable $HERDING_{k,t}$ and $EXPERIENCE_{i,t}$.

After incorporating all these variables to test the three suspected drivers of herding behavior within the ETF market, our base regression takes the following form:

$$HERDING_{k,t} = f(LIQ_{j,t-1}; LN_MKTSIZE_{j,t-1}; EXPERIENCE_{i,t}; \quad (7)$$

$$SMART_MONEY_{k,t-1}; Dummy_SMART_MONEY_{k,t-1}^* \\ SMART_MONEY_{k,t-1}; VARIOUS CONTROL VARIABLES)$$

We include multiple control variables in Equation (7), like previous quarter ETF returns, institutional investor type dummies, and ETF category dummies, to insure that our variables of interests stand a better change of capturing the true significance of the three factors of interests in our regressions. Table 6 presents a detailed description of all the variables used in the regression.

[Insert Table 6 Here]

5.4.4 Regression Results

Table 7 presents the results of six different specifications of the base regression in Equation (7) above. For the most part, these results lend strong support to our hypotheses that common preference/aversion to certain ETF characteristics and that institutional investors' experience are both significant drivers of institutional investor herding behavior in the ETF market.

That common preference/aversion to certain ETF characteristics is an important determinant of herding behavior is evidenced by the significantly negative coefficient on $LIQ_{j,t-1}$ and $LN_MKTSIZE_{j,t-1}$ across the six model specifications tested. It seems that institutional investors have an aversion for information uncertainty concerning ETF return prospects, and that they regard new trades on illiquid and small ETFs as a signal of new information arrival about those ETFs and trade accordingly. Consequently, herding is more pervasive amongst relatively illiquid and small ETFs.

That institutional investor experience in the ETF market matters is also evidenced by the significantly negative coefficient on $EXPERIENCE_{j,t-1}$ across the four model specifications where this variable was used as a regressor. This suggests that the longer the exposure institutional investors have to the ETF market, the less prone they are to herd.

In contrast, the hypothesis that institutional investors herd due to correlated information about ETF prospects does not obtain the same support. As Table 7 shows, the coefficient on $SMART_MONEY_{k,t}$, in spite of being of the expected sign, is consistently insignificant throughout the different model specifications tested. Even for the modified version of this variable, the interaction of $SMART_MONEY_{k,t}$ with $Dummy_SMART_MONEY_{k,t-1}$, a variable that equals to one when $SMART_MONEY_{k,t}$ is greater than zero (and zero otherwise), turns out

insignificant. This result is not particularly surprising given our earlier findings that institutional investors do not seem to successfully take strategic positions in ETF for profit, which may be due to their lack of valuable private information. However, we are also open to the possibility that $SMART_MONEY_{k,t}$ may not be a good proxy for private information in the hands of institutional investors; so we qualify our conclusions.

Although not tabulated for the sake of space, we also find that three ETF market type dummies used in the regression to control for differences in herding across ETF market types do not load significantly in either one of the six model specifications. In contrast, the three institutional investor type dummies used in the regression to control for differences in herding across investors types load significantly positive in all the six model specifications ran. This result suggests that herding is most pervasive when measured within institutional investors of the same kind. This is, banks are more likely to herd with other banks, insurance companies with insurance companies, and so forth.

[Insert Table 7 Here]

6. Conclusion

This paper examines institutional investor behavior in the ETF market by looking at institutional investor share ownership in a total of 367 ETFs operating from 1993 to 2007. To ensure that the sample of institutional investors trading in these ETF is large enough, we focus our attention to ETFs within the U.S. broad market, style, U.S. sector, and international market categories.

Based on Pearson correlations and on the performance measures of Grinblatt and Titman (1993) computed for our sample, we find that institutional investors do not seem to be able to successfully take strategic positions in the ETF market to profit directly from subsequent price

movements. Rather, the significantly negative performance measures found in our tests suggest that institutional investors use ETFs as a transient venue to help them comply their investment policy guidelines when they cannot do so with individual securities.

With respect to institutional investors' usage of feedback investment strategies within the ETF market, contrary to the gross of the evidence uncovered from institutional trading in the stock market, we find evidence that institutional investors engage in negative feedback (or contrarian) strategies. Although we do not test whether these contrarian strategies are effective in generating abnormal returns, the pattern is consistent with the idea that institutions see the ETF market as a viable means to trade on the basis of potential market overreaction to new information. Additional evidence of selective feedback trading across distinct ETF market categories also suggests that, to an extent, institutions also use ETFs for diversification purposes.

Furthermore, after controlling for the number of active institutional investors in an ETF, we find that institutional investors typically tend to herd when trading in the ETF market. Moreover, our tests find that the level of herding increases as the number of active institutional investors increases, and that herding is most pervasive amongst banks.

Through regression analysis, we find that herding behavior is affected by certain ETF traits and by the institutional investors' experience within the ETF market, rather than by correlated private information amongst investors. Since herding is most pervasive within relatively illiquid and small ETFs, we believe that institutional investors have a particular aversion for information uncertainty, regarding any trade in these ETFs as a sign of new information arrival and following with similar trades. That is, a buy will be followed by further buys and a sell will be followed by further sells. With respect to the role of experience in herding

behavior, regression results suggests that herding takes place mostly amongst institutional investors who are relatively inexperienced in the ETF market. As institutional investors get more years of ETF trading experience under their belt, they seem to rely less and less on herding.

References

- Bollen, Nicolas P. B. and Jeffrey A. Busse, 2001, "On the timing ability of mutual fund managers," *Journal of Finance*, 1075-1094.
- Bollen, Nicolas P. B. and Jeffrey A. Busse, 2005, "Short-term persistence in mutual fund performance," *Review of Financial Studies*, 569-597.
- Badrinath, S. G. and Sunil Wahal, 2002, "Momentum trading by institutions," *Journal of Finance*, 2449-2478.
- Banerjee, Abhijit V., 1992, "A simple model of herd behavior," *The Quarterly Journal of Economics*, 797-817.
- Bikhchandani, Sushil, David Hirshleifer, and Ivo Welch, 1992, "A Theory of Fads, Fashion, Custom, and Cultural Change as Informational Cascades," *The Journal of Political Economy*, 992-1026.
- Chan, Louis K.C., Yasushi Hamao, and Josef Lakonishok, 1991, "Fundamentals and stock returns in Japan," *Journal of Finance*, 1739-1764.
- Chancea, Don M. and Michael L. Hemlerb, 2001, "The performance of professional market timers: daily evidence from executed strategies," *Journal of Financial Economics*, 377-411.
- Chakravarty, Sugato, 2001, "Stealth-trading: Which traders' trades move stock prices?" *Journal of Financial Economics*, 289-307.
- Chevalier, J., and G. Ellison, 1999, "Career concerns of mutual fund managers," *Quarterly Journal of Economics* 114, 389-432.
- Falkenstein, E., 1996, "Preferences for stock characteristics as revealed by mutual fund portfolio holdings," *Journal of Finance*, 111-135.
- Froot, K., D. Scharfstein, and J. Stein, 1992, "Herd on the street: Informational efficiencies in a market with short-term speculation," *Journal of Finance*, 1461-1484.
- Gompers, Paul A. and Andrew Metrick, 2001, "Institutional investors and equity prices," *Quarterly Journal of Economics*, 229-259.
- Griffin, John M., Jeffrey H. Harris, and Selim Topaloglu, 2003, "The dynamics of institutional and individual trading," *Journal of Finance*, 2285-2320.
- Grinblatt, Mark and Sheridan Titman, 1993, "Performance measurement without benchmarks: An examination of mutual fund returns," *Journal of Business*, 47-68.
- Grinblatt, Mark, Sheridan Titman, and Russ Wermers, 1995, "Momentum investment strategies, portfolio performance, and herding: A study of mutual fund behavior," *American*

Economic Review, 1088-1105.

Kent, Daniel, Mark Grinblatt, Sheridan Titman, and Russ Wermers, 1997, "Measuring mutual fund performance with characteristic-based benchmarks," *Journal of Finance*, 1035-1069.

Kraus , Alan, and Hans R. Stoll, 1972, "Parallel Trading by Institutional Investors," *The Journal of Financial and Quantitative Analysis*, 2107-2138.

Lakonishok, Josef, Andrei Shleifer, and Robert W. Vishny, 1992, "The impact of institutional trading on stock prices," *Journal of Financial Economics*, 23-43.

Nofsinger, John and Richard Sias, 1999, "Herding and feedback trading by institutional and individual investors," *Journal of Finance*, 2263-2295.

Scharfstein, David S. and Jeremy C. Stein, 1990, "Herd Behavior and Investment," *The American Economic Review*, 465-479

Shiller, Robert J. and John Pound, 1989, "Survey evidence on diffusion of interest and information among investors," *Journal of Economic Behavior & Organization*, 47-66

Tse, Yiuman and Valeria Martinez, 2007, "Price discovery and informational efficiency of international iShares funds," *Global Finance Journal*, 1-15.

Wermers, Russ, 1999, "Mutual fund herding and the impact on stock prices," *Journal of Finance*, 581-622.

Wermers, Russ, 2000, "Mutual fund performance: an empirical decomposition into stock-picking talent, style, transaction costs, and expense," *Journal of Finance*, 1655-1695.

Zheng, Lu, 1999, "Is money smart? A study of mutual fund investor's fund selection ability," *Journal of Finance*, 901-903.

Table 1

Distribution of Institutional Ownership by ETF Category

Mean, median, standard deviation, minimum, and maximum percentage ownership of top institutional investors by ETF category. The institutional investor types correspond to the top four institutional investor categories provided by CDA/Spectrum's Institutional Holding database, which includes banks, insurance companies, investment companies, and independent investment advisors. The percentage ownership figures shown are calculated using the total stake held by these four investor types in each ETF category as the basis. A fifth category labeled "Others" is ignored due to a lack of information on its constitution. Panels A through D present the summary statistics for the four largest ETF investment objective categories by market capitalization standards: U.S. Broad Market, Style, U.S. Sector, and International ETFs, respectively. Summary statistics are reported for the 15 years in the sample, as well as for two subsample periods: one from 1993 to 2000 and one from 2001 to 2007.

Panel A: U.S. broad market		1993 - 2007					1993 - 2000					2001 - 2007				
		Mean	Median	STD	Min	Max	Mean	Median	STD	Min	Max	Mean	Median	STD	Min	Max
Banks		0.41	0.40	0.18	0.07	0.82	0.38	0.40	0.20	0.07	0.81	0.44	0.41	0.15	0.22	0.75
Insurance Companies		0.08	0.05	0.08	0.01	0.40	0.13	0.10	0.09	0.02	0.40	0.03	0.02	0.02	0.01	0.08
Investment Companies		0.02	0.01	0.03	0.00	0.14	0.03	0.01	0.04	0.00	0.14	0.01	0.00	0.01	0.00	0.04
Ind. Investment Advisors		0.50	0.53	0.20	0.09	0.83	0.48	0.47	0.22	0.09	0.83	0.53	0.56	0.15	0.21	0.76

Panel B: Style market		1993 - 2007					1993 - 2000					2001 - 2007				
		Mean	Median	STD	Min	Max	Mean	Median	STD	Min	Max	Mean	Median	STD	Min	Max
Banks		0.54	0.65	0.25	0.09	0.96	0.35	0.23	0.28	0.09	0.96	0.68	0.68	0.08	0.49	0.83
Insurance Companies		0.15	0.07	0.22	0.00	1.00	0.27	0.19	0.29	0.00	1.00	0.06	0.07	0.03	0.01	0.12
Investment Companies		0.04	0.01	0.08	0.00	0.42	0.14	0.06	0.20	0.00	0.42	0.02	0.01	0.03	0.00	0.09
Ind. Investment Advisors		0.33	0.27	0.20	0.02	0.86	0.45	0.50	0.25	0.02	0.86	0.24	0.25	0.07	0.11	0.40

Table 1 (Cont.)
Summary Statistics of Institutional Investors' Percentage Stake Ownership in ETFs

Panel C:	U.S. sector market	1993 - 2007						1993 - 2000						2001 - 2007					
		Mean	Median	STD	Min	Max	Mean	Median	STD	Min	Max	Mean	Median	STD	Min	Max	Mean	Median	STD
Banks		0.44	0.52	0.20	0.03	0.73	0.23	0.22	0.18	0.03	0.56	0.50	0.56	0.16	0.13	0.73			
Insurance Companies		0.09	0.03	0.18	0.00	0.90	0.29	0.17	0.27	0.05	0.90	0.02	0.01	0.03	0.00	0.12			
Investment Companies		0.00	0.00	0.00	0.00	0.00	-	-	-	-	-	0.00	0.00	0.00	0.00	0.00			
Ind. Investment Advisors		0.48	0.42	0.19	0.10	0.92	0.50	0.42	0.28	0.10	0.92	0.48	0.42	0.16	0.26	0.81			

Panel D:	International market	1993 - 2007						1993 - 2000						2001 - 2007					
		Mean	Median	STD	Min	Max	Mean	Median	STD	Min	Max	Mean	Median	STD	Min	Max	Mean	Median	STD
Banks		0.39	0.41	0.41	0.02	0.70	0.03	0.04	0.01	0.02	0.04	0.43	0.42	0.16	0.13	0.70			
Insurance Companies		0.09	0.08	0.06	0.00	0.25	0.06	0.09	0.05	0.00	0.09	0.09	0.07	0.06	0.02	0.25			
Investment Companies		0.00	0.00	0.00	0.00	0.02	-	-	-	-	-	0.00	0.00	0.00	0.00	0.02			
Ind. Investment Advisors		0.53	0.51	0.19	0.14	0.96	0.91	0.88	0.05	0.87	0.96	0.48	0.50	0.15	0.14	0.74			

Table 2
Correlation Between Institutional Investment Weight
Changes and ETF Returns

Pearson correlation coefficient between institutional investors' quarterly investment weight changes and ETF returns the following month. The correlation coefficients are followed by their respective probability values in parentheses. Symbols *, **, and *** denote significance of correlation coefficients at the 10, 5 and 1 % levels, respectively. Panel A reports correlation coefficients for all institutional investors, and Panels B through E report correlation coefficients amongst the top four institutional investor categories identified by CDA/Spectrum. Each panel reports the correlation coefficients achieved by these institutional investor categories within the four top ETF market categories, by market capitalization standards. The table reports summary statistics for these investor categories using the full 15 years in the sample, as well as two subsample periods: one from 1993 to 2000 and one from 2001 to 2007. Where no correlation estimates are reported, insufficient institutional investor holdings prevents the computation of quarterly weight differences.

Panel A: All Institutional Investors			
	1993-2007	1993-2000	2001-2007
U.S. broad market	0.001 (0.57)	0.000 (0.67)	0.004 (0.07) *
Style market	0.000 (0.75)	0.005 (0.27)	0.000 (0.95)
U.S. sector market	0.001 (0.70)	-0.001 (0.91)	0.027 (0.30)
International market	0.003 (0.89)	-0.010 (0.69)	0.001 (0.76)
Panel B: Banks			
	1993-2007	1993-2000	2001-2007
U.S. broad market	0.001 (0.40)	0.001 (0.74)	0.002 (0.10) *
Style market	0.002 (0.10) *	0.010 (0.08)	0.002 (0.13)
U.S. sector market	0.002 (0.15)	0.009 (0.34)	0.022 (0.13)
International market	0.003 (0.03) **	0.009 (0.54)	0.003 (0.05)
Panel C: Insurance Companies			
	1993-2007	1993-2000	2001-2007
U.S. broad market	0.010 (0.52)	-0.015 (0.06)	0.040 (0.17)
Style market	-0.002 (0.82)	-0.025 (0.16)	0.025 (0.38)
U.S. sector market	0.015 (0.27)	-0.037 (0.40)	0.037 (0.21)
International market	0.036 (0.32)	-0.010 (-)	0.037 (0.32)
Panel D: Investment Companies and Their Managers			
	1993-2007	1993-2000	2001-2007
U.S. broad market	-0.001 (0.35)	-0.014 (0.37)	0.000 (0.66)
Style market	0.000 (0.79)	-	0.000 (0.68)
U.S. sector market	0.004 (0.68)	-	0.004 (0.68)
International market	0.002 (0.48)	-	0.002 (0.50)
Panel E: Independent Investment Advisors			
	1993-2007	1993-2000	2001-2007
U.S. broad market	0.000 (0.91)	0.001 (0.51)	0.002 (0.54)
Style market	-0.003 (0.28)	0.005 (0.47)	-0.004 (0.20)
U.S. sector market	-0.002 (0.71)	-0.010 (0.72)	0.000 (0.99)
International market	-0.005 (0.07) *	-0.035 (0.63)	-0.004 (0.13)

Table 3
Institutional Investors' Performance Estimates

Average institutional investor performance estimates using Grinblatt and Titman (1993) methodology. The performance measure for institutional investor i , which we term *ETF Change Measure_i*, is given by:

$$ETF\ Change\ Measure_i = \sum_{t=1}^T \sum_{j=1}^N [R_{j,t} (w_{j,t}^i - w_{j,t-1}^i)] / T,$$

where $R_{j,t}$ is ETF j 's return in the month immediately following Quarter t and $w_{j,t}^i$ is the weight of institutional investor i 's position in ETF j in Quarter t . These performance measures are averaged across all the institutional investors in the subsample of interest. N reports the number of ETF Change Measures used in the calculation of these averages. t -statistics are computed dividing the mean performance estimate in that subgroup by the subgroups' mean's standard error. Symbols *, **, and *** denote significance of means at the 10, 5 and 1 % levels, respectively. Panel A reports average performance for all institutional investors, and Panels B through E report average performance measures for the top four institutional investor categories identified by CDA/Spectrum. Each panel reports the average performance achieved by these institutional investor categories in the four top ETF market categories, by market capitalization standards. The table reports average performance measures using data from the full 15 years in the sample, as well as for two subsample periods: one from 1993 to 2000 and one from 2001 to 2007. Where no statistics are reported, insufficient institutional investor holdings prevents the computation of quarterly weight differences. Where N is reported to be one, no average is computed and the ETF change measure for that sole institutional investor in that ETF category is reported.

Panel A: All Institutional Investors

	1993-2007			1993-2000			2001-2007		
	N	Mean	t-stat.	N	mean	t-stat.	N	mean	t-stat.
U.S. broad market	447	-0.111***	-3.11	211	-0.182***	-3.08	311	-0.163***	-3.24
Style market	277	-0.143***	-2.79	65	-0.289**	-2.16	240	-0.166***	-2.81
U.S. sector market	219	-0.198***	-2.85	20	-0.879**	-2.13	212	-0.218***	-3.07
International market	248	-0.173***	-2.71	10	-0.635	-0.78	247	-0.189***	-3.06

Panel B: Banks

	1993-2007			1993-2000			2001-2007		
	N	mean	t-stat.	N	mean	t-stat.	N	mean	t-stat.
U.S. broad market	171	-0.142***	-2.56	91	-0.250***	-2.53	135	-0.180***	-2.58
Style market	127	-0.192***	-2.59	32	-0.206	-1.41	116	-0.209***	-2.58
U.S. sector market	108	-0.224***	-2.57	12	-0.613*	-1.72	103	-0.239***	-2.62
International market	102	-0.259***	-2.94	5	-0.155	-1.34	102	-0.259***	-2.94

Panel C: Insurance companies

	1993-2007			1993-2000			2001-2007		
	N	mean	t-stat.	N	mean	t-stat.	N	mean	t-stat.
U.S. broad market	28	-0.362	-1.65	18	-0.254	-1.13	16	-0.637	-1.72
Style market	12	-0.150*	-1.74	6	-0.103	-0.63	9	-0.207***	-2.60
U.S. sector market	9	-0.710	-1.38	2	-0.447***	-37.35	9	-0.710	-1.38
International market	11	-0.849	-1.59	1	-0.435	-	11	-0.849	-1.59

Panel D: Investment companies and Their Managers

	1993-2007			1993-2000			2001-2007		
	N	Mean	t-stat.	N	Mean	t-stat.	N	mean	t-stat.
U.S. broad market	10	0.018	0.75	7	0.022	0.65	4	-0.012	-0.38
Style market	4	-0.376	-1.04	-	-	-	4	-0.376	-1.04
U.S. sector market	2	0.032	0.79	-	-	-	2	0.032	-0.79
International market	4	-0.014	-0.45	-	-	-	4	-0.014	-0.45

Panel E: Independent Investment Advisors

	1993-2007			1993-2000			2001-2007		
	N	mean	t-stat.	N	mean	t-stat.	N	mean	t-stat.
U.S. broad market	238	-0.064	-1.37	95	-0.120	-1.47	156	-0.104	-1.48
Style market	134	-0.089	-1.13	27	-0.429	-1.58	111	-0.109	-1.16
U.S. sector market	100	-0.129	-1.16	6	-1.556	-1.28	98	-0.157	-1.41
International market	131	-0.054	-0.62	4	-1.285	-0.60	130	-0.084	-1.02

Table 4
ETF Performance and Institutional Investor Demand

Quarterly institutional investor excess demand measures, averaged across previous-quarter performance quintiles and market capitalization quintiles. Panel A presents average dollar excess demand measures (*Dratios*) and Panel B presents average number of institutional investors increasing their ETF holdings in the quarter (*Nratios*), both used in Lakonishok, et al. (1992). To be more precise, $Dratio_{j,t}$ is defined as $[\$buys_{j,t} - \$sells_{j,t}] / [\$buys_{j,t} + \$sells_{j,t}]$, where $\$buys_{j,t}$ is the total dollar increases by all institutional investors in ETF j in Quarter t (evaluated at the average ETF price during the quarter) and $\$sells_{j,t}$ is the total dollar decreases by all institutional investors in ETF j in Quarter t (evaluated at the average ETF price during the quarter). $Nratio_{j,t}$ is defined as $\#buys_{j,t} / \#active_{j,t}$, where $\#buys_{j,t}$ equals the number of institutional investors increasing their holdings of ETF j in Quarter t and $\#active_{j,t}$ is the number of institutional investors changing their holdings of ETF j in Quarter t . To ensure that each ETF cohort is comprised of at least five ETFs, average excess demand measures are only computed using ETF data from 2001 to 2007. Similarly, to prevent these measures from being biased by one or two institutional investors, all ETFs in the table's cohorts must be traded by at least five institutional investors. Past-quarter performance quintiles are presented in ascending performance order, with the first quintile (1) representing the worst performing ETFs in the previous quarter, and the last quintile (5) representing the best performing ETFs in the previous quarter. Similarly, size quintiles are presented in increasing order, with the first quintile (1) representing the smallest ETFs in the sample that quarter, and the last quintile (5) representing the largest ETFs in sample that quarter. t-statistics are reported in parentheses under the average excess dollar demand ratios in panel A, while standard deviations are reported in parentheses in Panel B, under the average proportion of institutional investors buying ETF in the quarter. Symbols *, **, and *** in Panel A denote significance of means at the 10, 5 and 1 % levels, respectively.

Past-quarter performance	Size				
	1 (small)	2	3	4	5 (large)
Panel A: Excess dollar demand ratio (D-ratio)					
1 (worst)	0.195*** (4.69)	0.070*** (2.10)	0.037 (1.00)	-0.011 (-0.32)	0.131*** (3.05)
2	0.177*** (3.81)	0.114** (2.62)	0.009 (0.18)	0.070** (2.14)	-0.006 (-0.25)
3	0.042 (0.70)	-0.015 (-0.33)	-0.031 (-0.91)	0.013 (0.40)	-0.023 (-0.85)
4	0.068 (1.38)	0.009 (0.23)	-0.019 (-0.49)	-0.070** (-2.23)	-0.080*** (-2.83)
5 (best)	0.018 (0.40)	-0.057 (-1.40)	0.006 (0.18)	0.046 (1.33)	-0.100** (-2.27)
Panel B: Proportion of institutional investors buying (N-ratio)					
1 (worst)	0.605 (0.180)	0.535 (0.149)	0.530 (0.142)	0.504 (0.124)	0.571 (0.148)
2	0.585 (0.189)	0.554 (0.172)	0.520 (0.181)	0.535 (0.143)	0.492 (0.111)
3	0.536 (0.201)	0.488 (0.183)	0.488 (0.154)	0.508 (0.141)	0.495 (0.122)
4	0.549 (0.182)	0.512 (0.160)	0.493 (0.150)	0.472 (0.128)	0.460 (0.124)
5 (best)	0.515 (0.208)	0.472 (0.183)	0.502 (0.142)	0.523 (0.141)	0.450 (0.141)

Table 5
Institutional Investors Herding in the ETF Market

Average institutional investor herding in the ETF market from 1993 to 2007, using the herding measure introduced by Lakonishok, et al. (1992). The herding measure for ETF j in Quarter t , is given by:

$$H_{j,t} = \left| \frac{B_{j,t}}{B_{j,t} + S_{j,t}} - P_{j,t} \right| - AF_{j,t},$$

where $B_{j,t}$ is the number of institutional investors who increase their holdings in ETF j in Quarter t (net buyers), $S_{j,t}$ is the number of institutional investors who decrease their holdings in ETF j in Quarter t (net sellers), $P_{j,t}$ is the expected proportion of institutional investors buying ETF j in Quarter t relative to the number of active institutional investors in that ETF that quarter, and $AF_{j,t}$ is an adjustment factor equal to the expected value of the absolute term in $H_{j,t}$ above, under the null hypothesis of no herding. Herding figures are averaged over all the ETFs held by a the institutional investor type of interest. Panel A reports average herding amongst all institutional investors in all ETF categories. Panel B reports average herding for all institutional investors, separated by the top four ETF categories according to market capitalization. Panels C and D respectively report average herding amongst banks and independent investment advisors, separated by the top four ETF categories according to market capitalization. Due to the low number of insurance and investment companies actively trading in the ETF market throughout the sample period (see Panels C and D in Table 3), it is not possible to analyze herding this way for these two types of institutional investors. Column 1 reports mean and median statistics of herding behavior within all the ETFs in the sub-sample of interest, regardless of the number of institutional investors actively trading (i.e. buying and selling) these ETFs. Columns 2 and 3 focus only on herding behavior within ETFs with at least five and ten active institutional investors, respectively. Column 4 reports the difference between the herding measures in Columns 3 and 1. N reports the number ETF-quarters used in the calculation of these averages. P is the expected proportion of institutional investors buying ETFs relative to the number of active institutional investors in that ETF that quarter. t-statistics are reported in parentheses below the means and computed by dividing the mean herding estimate in that subgroup by the subgroups' mean's standard error. Symbols *, **, and *** denote significance of means at the 10, 5 and 1% levels, respectively.

Panel A: Herding in the ETF market

	(1) All ETFs	(2) ETFs with more than 5 active institutions	(3) ETFs with more than 10 active institutions
Mean (H_i)	0.00	0.02***	0.02***
t-stat.	(-1.61)	(10.14)	(9.66)
Median	-0.02***	0.01	0.01***
Sign test (p-value)	(0.00)	(0.00)	(0.00)
N (ETF-quarter)	2,779	1,735	1,141

Panel B: Herding in the ETF market, by ETF investment objective

	(1) All ETFs			(2) ETFs with more than 5 active institutions			(3) ETFs with more than 10 active institutions			(4) Mean Difference (3)-(1)
	N	P	Mean	N	P	Mean	N	P	Mean	
U.S. Broad Market	341	0.56	0.00 (-0.24)	222	0.56	0.02*** (3.77)	185	0.56	0.02*** (3.48)	0.02*** (2.61)
Style Market	726	0.63	-0.01 (-2.02)	445	0.63	0.02*** (4.00)	349	0.63	0.01*** (3.87)	0.02*** (4.16)
U.S. Sector Market	967	0.57	0.00 (-0.48)	639	0.58	0.02*** (4.05)	357	0.57	0.01** (2.28)	0.01** (2.03)
International market	745	0.65	-0.01 (-1.57)	429	0.66	0.02*** (4.49)	250	0.66	0.02*** (4.15)	0.03*** (4.29)
Total	2,779			1,735			1,141			

Table 5 (Cont.)
Insitutional Investors Herding in the ETF Market

Panel C: Herding amongst banks in the ETF market, by ETF investment objective

	(1) All ETFs			(2) ETFs with more than 5 active banks			(3) ETFs with more than 10 active banks			(4) Mean Difference (3)-(1)
	N	P	Mean	N	P	Mean	N	P	Mean	
U.S. Broad Market	317	0.57	0.00 (0.43)	207	0.57	0.02*** (3.44)	170	0.57	0.03*** (4.37)	0.03*** (2.94)
Style Market	658	0.63	-0.01*** (-2.37)	385	0.63	0.02*** (4.39)	313	0.63	0.02*** (4.40)	0.03*** (4.82)
U.S. Sector Market	870	0.57	-0.01* (-1.72)	478	0.58	0.01*** (2.04)	258	0.57	0.01*** (2.17)	0.02*** (2.77)
International market	635	0.67	-0.02*** (-3.48)	317	0.67	0.01** (2.10)	135	0.67	0.03*** (3.46)	0.05*** (4.15)
Total	2,480			1,387			876			

Panel D: Herding amongst independent investment advisors in the ETF market, by ETF investment objective

	(1) All ETFs			(2) ETFs with more than 5 active advisors			(3) ETFs with more than 10 active advisors			(4) Mean Difference (3)-(1)
	N	P	Mean	N	P	Mean	N	P	Mean	
U.S. Broad Market	239	0.55	-0.02*** (-2.44)	127	0.55	0.01 (0.80)	83	0.55	0.00 (0.20)	0.02* (1.79)
Style Market	554	0.62	-0.02*** (-4.31)	215	0.62	0.02** (2.50)	77	0.60	0.02** (2.00)	0.05*** (3.14)
U.S. Sector Market	703	0.58	-0.04*** (-8.35)	145	0.56	-0.01 (-0.62)	41	0.54	0.00 (0.42)	0.04*** (3.46)
International market	573	0.63	-0.04*** (-9.24)	149	0.64	0.01 (1.16)	58	0.63	0.02 (1.53)	0.06*** (4.00)
Total	2,069			636			259			

Table 6
Variables Used in the Herding Behavior Regressions

Glossary of all the variables used in the herding regression depicted in Equation 7. The variable name is presented in the first column, a detailed definition is provided in the second column, and the last column shows the hypothesis tested with the addition of that variable to the regression.

Variable	Definition	Hypothesis Tested
LN_PRC/100	LN(ETF price in the Previous Quarter)/100 (Falkenstein, 1996)	Control
STDEV/100	(Standard deviation of ETF monthly returns over the trailing period starting 18 months ago to 6 months ago) / 100 (Falkenstein, 1996)	Control
LIQ/100	[Trading volume of ETFs in the prior quarter / ETF Outstanding shares in the prior quarter]/100 (Falkenstein, 1996)	Shared aversion to certain ETF characteristics
LN_MKTSIZE/100	LN(ETF Price in the prior quarter * ETF Outstanding Shares in the prior quarter) / 100 (Falkenstein, 1996)	Shared aversion to certain ETF characteristics
EXPERIENCE	Number of years institutional investor has been trading ETFs (Chevalier and Ellison, 1999)	Role of investor experience in herding
SMART_MONEY	Grinblatt and Titman's (1993) performance component averaged across all institutional investor of type k in the sample (Grinblatt and Titman, 1993)	Presence of correlated private information
Dummy_SM	If SMART_MONEY is larger than 0, Dummy_SM is equal to 1. Otherwise, Dummy_SM is equal to 0.	Not used by itself
Dummy_SM* SMART_MONEY	Dummy_SM*SMART_MONEY is equal to SMART_MONEY multiplied by Dummy_SM.	Presence of correlated private information
QRET/100	Average monthly ETF returns over the previous quarter	Control
DUMMY_BANK	DUMMY_BANK is equal to 1 if the investor is a bank. Otherwise, DUMMY_BANK is equal to 0.	Control
DUMMY_INS	DUMMY_INS is equal to 1 if the investor is an insurance company. Otherwise, DUMMY_INS is equal to 0.	Control
DUMMY_IIA	DUMMY_IIA is equal to 1 if the investor is an independent investment advisor. Otherwise, DUMMY_IIA is equal to 0.	Control
DUMMY_STYLE	DUMMY_STYLE is equal to 1 for ETFs within the style ETF category. Otherwise, DUMMY_STYLE is equal to 0.	Control
DUMMY_SECTOR	DUMMY_SECTOR is equal to 1 for ETFs within the U.S. sector ETF category. Otherwise, DUMMY_SECTOR is equal to 0.	Control
DUMMY_IM	DUMMY_IM is equal to 1 for ETFs within the international markets ETF category. Otherwise, DUMMY_IM is equal to 0.	Control

Table 7
Determinants of Herding Behavior

Result of regressions exploring the determinants of herding behavior amongst institutional investors in the ETF market. The dependent variable is the level of herding observed amongst institutional investors of type k in Quarter t , defined as:

$$HERDING_{k,t} = \sum_{i=1}^{I(k)} \sum_{j=1}^N w_{j,t}^i H_{j,t} ,$$

where $H_{j,t}$ is the herding measure for ETF j in Quarter t , as defined in Equation 4, $w_{j,t}^i$ is the weight of institutional investor i 's position in ETF j in Quarter t , and $I(k)$ is the total number of institutional investors of type k in the sample. Six variations on the following base regression are reported:

$$HERDING_{k,t} = f(LIQ_{j,t-1}; LN_MKTSIZE_{j,t-1}; EXPERIENCE_{t,t}; SMART_MONEY_{k,t-1}; Dummy_SMART_MONEY_{k,t-1} * SMART_MONEY_{k,t-1}; VARIOUS CONTROLS)$$

A blank space appears instead of a coefficient whenever a variable is not used in a particular model specification, except in the case of the three investor type dummies and the three ETF market category dummies, all of which were used in the six model specifications below. Although their coefficients are not reported for the sake of economizing space, the coefficients on the three investor type dummies loaded significantly across all six models. In contrast, the three ETF market category dummies loaded significantly across all six model specifications. Probability values are included in parentheses below the corresponding reported coefficient. Symbols *, **, and *** denote significance of coefficients at the 10, 5 and 1% levels, respectively. The number of ETF-quarter observations in the sample used in these regressions (N) and measures of goodness of fit of the regressions (R^2 and Adj- R^2) can be found in the three bottom rows.

	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
Intercept	0.0769 *** (<.0001)	0.1021 *** (<.0001)	0.07744 *** (<.0001)	0.10284 *** (<.0001)	0.09874 *** (<.0001)	0.0994 *** (<.0001)
LN_PRC/100	-0.15283 (0.2988)	0.17895 (0.2378)	-0.168 (0.2518)	0.168 (0.2672)	0.09959 (0.5191)	0.0872 (0.5719)
STDEV /100	-0.01295 (0.7535)	0.03091 (0.4418)	0.00155 (0.9698)	0.04531 (0.2550)	0.00555 (0.8931)	0.01941 (0.6354)
LIQ /100	-0.00825 ** (0.0321)	-0.00881 ** (0.0215)	-0.00842 ** (0.0287)	-0.00899 ** (0.0191)	-0.00807 ** (0.0357)	-0.00823 ** (0.0322)
LN_MKTSIZE/100		-0.56505 *** (<.0001)		-0.57118 *** (<.0001)	-0.42506 *** (<.0001)	-0.42823 *** (<.0001)
EXPERIENCE	-0.0032 *** (<.0001)		-0.00325 *** (<.0001)		-0.0014 *** (0.0083)	-0.00143 *** (0.0071)
SMART_MONEY	-3.46767 (0.1012)	-3.26472 (0.1222)			-3.19543 (0.1301)	
Dummy_SM*SMART_MONEY			-7.8501 (0.1231)	-7.54821 (0.1374)		-7.87794 (0.1209)
QRET /100	-0.03653 (0.2079)	-0.04498 (0.1196)	-0.03617 (0.2125)	-0.04477 (0.1213)	-0.0399 (0.1680)	-0.03958 (0.1714)
N	5,255	5,255	5,255	5,255	5,255	5,255
R ²	0.0314	0.0351	0.0313	0.0351	0.0364	0.0364
Adj. R ²	0.0292	0.0329	0.0291	0.0329	0.0340	0.0340

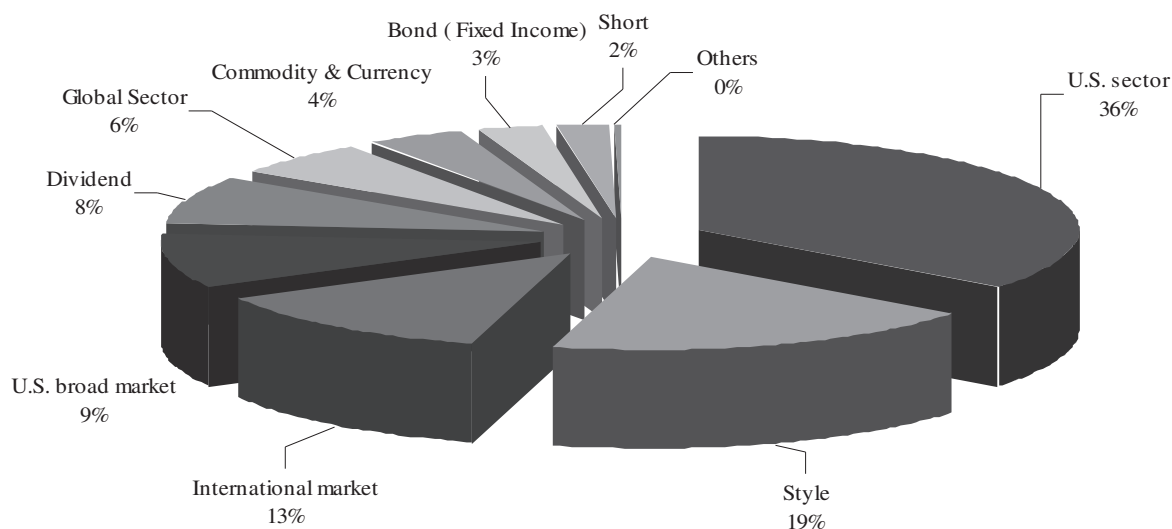


Figure 1
The Constitution of the ETF Market

Constitution of the ETF market as of the end of 2007, using AMEX and Morningstar's ETF investment objective taxonomy. The percentage figures represent the different ETF categories' weight in the overall ETF market, by market capitalization. The U.S. sector category includes ETFs that follow industry sector indices, like consumer staples, energy, utilities, and information technology indices. The style group mainly tracks U.S. growth, value, small-, middle-, and large-cap indices. The international market group tracks foreign broad market indices, including those of emerging markets like Brazil, Russia, India, and China. The U.S. broad market category includes ETFs that track major U.S. market indices, like the S&P 500 and Dow Jones indices. The dividend group contains ETFs that try to replicate indices of high dividend-yield stocks. The global sector group does the same as the U.S. sector group, but track indices that include foreign securities within the different industry sectors, not just U.S. securities. As the name suggest, commodity and currency ETFs try to replicate indices tracking the performance of commodity and currency indices, respectively. ETF in the bond category try to replicate the performance of diverse fixed income indices, like government, corporate, or municipal bond indices. Within the short ETF category you will find funds trying to provide the opposite performance as their benchmark index. Finally, the other category only includes one ETF, First Trust's U.S. IPO Index ETF (FPX), which tracks the performance of the IPOX-100 U.S. index.

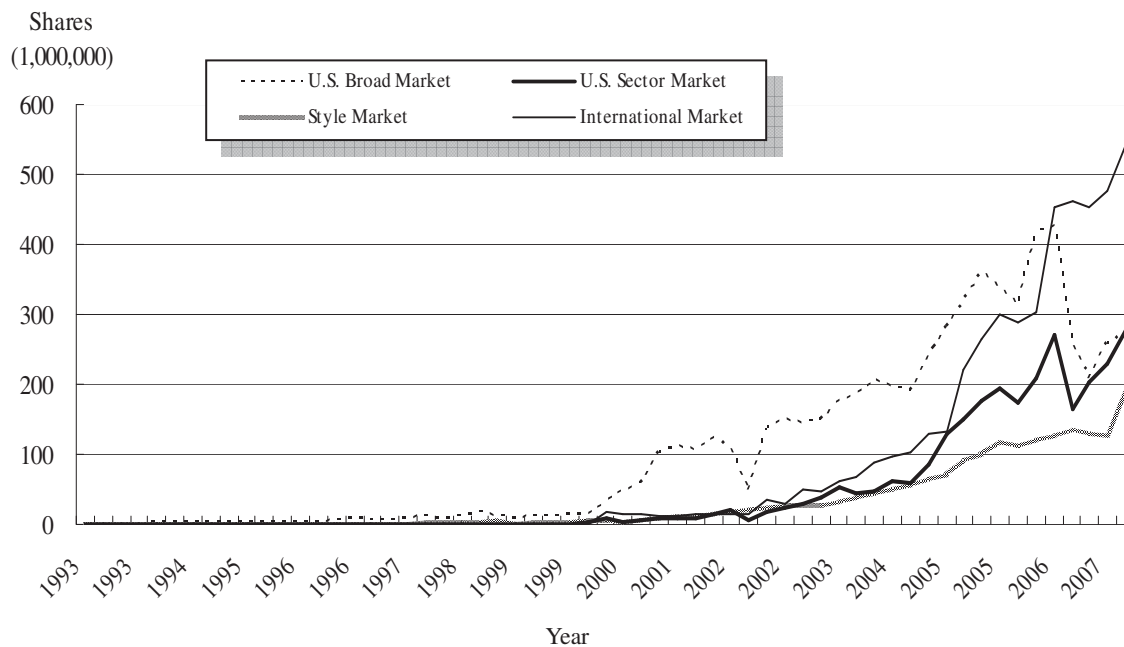


Figure 2

Shares held by Institutional Investors in the Top Four ETF Groups

Number of shares held (in millions) by institutional investors in the top four ETF groups (by market capitalization standards) from the birth of the ETF market in January 1993 to December 2007.

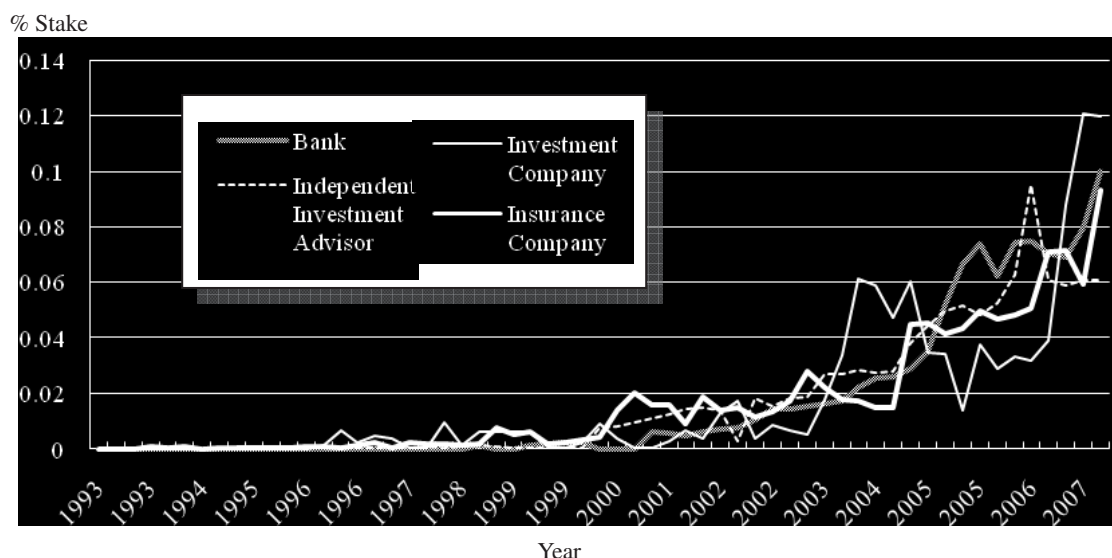


Figure 3
Institutional Investors' stake in the ETF Market

Percentage stake held by the four largest institutional investor types in the top four ETF groups, from the birth of the ETF market in January 1993 to December 2007. The institutional investor types depicted correspond to the top four institutional investor categories provided by CDA/Spectrum's Institutional Holding database. The top four ETF groups include U.S. sector, style, international market, and U.S. broad market ETF categories, the four largest ETF categories by market capitalization standards. The percentage stakes reported are calculated by adding the total number of ETF shares held each year in the top four ETF categories by each one of the institutional investor types depicted in the figure and dividing them by the total number of shares outstanding in the the top four ETF categories.

Appendix A

Computing Adjusted Factor $AF_{j,t}$ in the Herding Measure

We next show how to compute the adjusted factor, $AF_{j,t}$, of the herding measure in Equation 4. Recall that $AF_{j,t}$ is the expected value of $|B_{j,t}/(B_{j,t} - S_{j,t}) - P_{j,t}|$ under the null hypothesis of no herding and is subtracted from the first element in Equation 4 to account for the fact that that $|B_{j,t}/(B_{j,t} - S_{j,t}) - P_{j,t}|$ will typically be greater than zero for all $P_{j,t} > 0$.

Since $P_{j,t}$ represents the proportion of institutional investors buying ETF j in Quarter t relative to the number of institutional investor active that quarter, $B_{j,t}$ can be said to follow a binomial distribution with probability $P_{j,t}$ of success (i.e. of an active investor being a buyer).

Table A1 provides a numerical example showing the computation of $AF_{j,t}$ when the number of active institutional investors fluctuates between two and 15 ($n = 2$ to 15) and $P_{j,t} = 0.6$. Panel A provides first the probability of $B_{j,t} = x$ (for $x = 0$ to 15). Recall that if a random variable X follows a binomial distribution with parameters n and p , the probability of getting exactly x successes in n trials is given by the probability mass function:

$$Pr(X=x) = \frac{n!}{x!(n-x)!} p^x (1-p)^{n-x} \quad (A1)$$

The figures in Panel A are obtained using this formula. Panel B uses the probability value $p = 0.60$ to compute the value of $|B_{j,t}/(B_{j,t} - S_{j,t}) - P_{j,t}|$ for $x = 0$ to 15 and $n = 2$ to 15. For instance, for $x = 3$ and $n = 10$, this element is computed as follows:

$$|x/n - p| = |3/10 - 0.6| = 0.30$$

Table A1 Computation of Adjustment Factor $AF_{j,t}$ when the number of active institutional investors fluctuates between two and 15 ($n = 2$ to 15) and $p_{j,t} = 0.6$.

Panel A. Probability mass function of a binomial distribution with parameters $n = 2$ to 15 and $p = 0.60$.														
x	Number of Active Investors (n)													
	2	3	4	5	6	7	8	9	10	11	12	13	14	15
0	0.160	0.064	0.026	0.010	0.004	0.002	0.001	0.000	0.000	0.000	0.000	0.000	0.000	0.000
1	0.480	0.288	0.154	0.077	0.037	0.017	0.008	0.004	0.002	0.001	0.000	0.000	0.000	0.000
2	0.360	0.432	0.346	0.230	0.138	0.077	0.041	0.021	0.011	0.005	0.002	0.001	0.001	0.000
3		0.216	0.346	0.346	0.276	0.194	0.124	0.074	0.042	0.023	0.012	0.006	0.003	0.002
4			0.130	0.259	0.311	0.290	0.232	0.167	0.111	0.070	0.042	0.024	0.014	0.007
5				0.078	0.187	0.261	0.279	0.251	0.201	0.147	0.101	0.066	0.041	0.024
6					0.047	0.131	0.209	0.251	0.251	0.221	0.177	0.131	0.092	0.061
7						0.028	0.090	0.161	0.215	0.236	0.227	0.197	0.157	0.118
8							0.017	0.060	0.121	0.177	0.213	0.221	0.207	0.177
9								0.010	0.040	0.089	0.142	0.184	0.207	0.207
10									0.006	0.027	0.064	0.111	0.155	0.186
11										0.004	0.017	0.045	0.085	0.127
12											0.002	0.011	0.032	0.063
13												0.001	0.007	0.022
14													0.001	0.005
15														0.000

Table A1 (Continued)

Panel B. Value of $ B_{j,t}/(B_{j,t} - S_{j,t}) - p_{j,t} $ for $n = 2$ to 15 and $x = 0$ to 15 when $p_{j,t} = 0.60$														
	Number of Active Investors (n)													
x	2	3	4	5	6	7	8	9	10	11	12	13	14	15
0	0.600	0.600	0.600	0.600	0.600	0.600	0.600	0.600	0.600	0.600	0.600	0.600	0.600	0.600
1	0.100	0.267	0.350	0.400	0.433	0.457	0.475	0.489	0.500	0.509	0.517	0.523	0.529	0.533
2	0.400	0.067	0.100	0.200	0.267	0.314	0.350	0.378	0.400	0.418	0.433	0.446	0.457	0.467
3		0.400	0.150	0.000	0.100	0.171	0.225	0.267	0.300	0.327	0.350	0.369	0.386	0.400
4			0.400	0.200	0.067	0.029	0.100	0.156	0.200	0.236	0.267	0.292	0.314	0.333
5				0.400	0.233	0.114	0.025	0.044	0.100	0.145	0.183	0.215	0.243	0.267
6					0.400	0.257	0.150	0.067	0.000	0.055	0.100	0.138	0.171	0.200
7						0.400	0.275	0.178	0.100	0.036	0.017	0.062	0.100	0.133
8							0.400	0.289	0.200	0.127	0.067	0.015	0.029	0.067
9								0.400	0.300	0.218	0.150	0.092	0.043	0.000
10									0.400	0.309	0.233	0.169	0.114	0.067
11										0.400	0.317	0.246	0.186	0.133
12											0.400	0.323	0.257	0.200
13												0.400	0.329	0.267
14													0.400	0.333
15														0.400

Panel C. Product of probability mass function is Panel A and the value of $ B_{j,t}/(B_{j,t} - S_{j,t}) - p_{j,t} $ in Panel B for $n = 2$ to 15 and $x = 0$ to 15 when $p_{j,t} = 0.60$														
	Number of Active Investors (n)													
x	2	3	4	5	6	7	8	9	10	11	12	13	14	15
0	0.096	0.038	0.015	0.006	0.002	0.001	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
1	0.048	0.077	0.054	0.031	0.016	0.008	0.004	0.002	0.001	0.000	0.000	0.000	0.000	0.000
2	0.144	0.029	0.035	0.046	0.037	0.024	0.014	0.008	0.004	0.002	0.001	0.001	0.000	0.000
3		0.086	0.052	0.000	0.028	0.033	0.028	0.020	0.013	0.008	0.004	0.002	0.001	0.001
4			0.052	0.052	0.021	0.008	0.023	0.026	0.022	0.017	0.011	0.007	0.004	0.002
5				0.031	0.044	0.030	0.007	0.011	0.020	0.021	0.018	0.014	0.010	0.007
6					0.019	0.034	0.031	0.017	0.000	0.012	0.018	0.018	0.016	0.012
7						0.011	0.025	0.029	0.021	0.009	0.004	0.012	0.016	0.016
8							0.007	0.017	0.024	0.023	0.014	0.003	0.006	0.012
9								0.004	0.012	0.019	0.021	0.017	0.009	0.000
10									0.002	0.008	0.015	0.019	0.018	0.012
11										0.001	0.006	0.011	0.016	0.017
12											0.001	0.004	0.008	0.013
13												0.001	0.002	0.006
14													0.000	0.002
15														0.000
$AF_{j,t}$	0.288	0.230	0.207	0.166	0.166	0.149	0.139	0.134	0.120	0.120	0.114	0.109	0.106	0.099

Similarly, Panel C simply shows the product of the values in Panel A and C for $x = 0$ to 15 and $n = 2$ to 15, providing the weighted values of $|B_{j,t}/(B_{j,t} - S_{j,t}) - P_{j,t}|$ that will be added to compute the expected value of $|B_{j,t}/(B_{j,t} - S_{j,t}) - P_{j,t}|$ under the null hypothesis of no herding, namely $AF_{j,t}$. The last row in Panel C actually presents the column sum of these values, that is, the value of the adjustment factor $AF_{j,t}$. As this last row shows, there is an inverse relation between $AF_{j,t}$ and n , the number of active investors in ETF j in quarter, as $AF_{j,t}$ fluctuates from 0.288 when $n = 2$ to 0.099 when $n = 15$.